AI ACADEMY · LEVEL 5 · GRADE 10 · WEEK 12 · STUDENT LESSON

Training a small network

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Image-pattern Network Investigation

THE CONNECTED WEEK

Freeze the hidden step. Train one weight. Record every state.

Trace a complete restricted-network training loop with honest update counts and evaluation limits.

Keep one habit: Full configuration → five-state/four-update ledger → pictured one-pair fit → reserved/negative-input checks → independent new training pair.

MAIN LESSON · SECTION 1

What actually changes during this training loop?

> Repeat a complete calculation while keeping frozen values and the update budget explicit.

Learning goals

I can trace repeated forward loss gradient and update steps.

I can keep **frozen parameter**s distinct from the trained parameter.

I can report **training** results with their evaluation limits.

Retrieve the saved-state update

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Updating parameters. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Slopes and local change. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Forward passes by hand. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

Prerequisite and starting response

- Why should all parameters in one simultaneous update use the same saved gradient?

> **My starting explanation**

End goal

By the end, I can explain **training** a small network using a traced example and a stated limitation.

▼ **Earlier wording, alternatives and picture transcripts**

Your learning target

Remember before reading

This week you will practise:

Trace repeated forward loss gradient and update steps

Keep **frozen parameter**s distinct from the trained parameter

Report **training** results with their evaluation limits

Retrieve what you already know

This week’s end goal

MAIN LESSON · SECTION 2

Train one output weight,not every network value

> The hidden computation and both biases are frozen throughout this experiment.

Complete model,data,objective,start,rate and budget

Use one hidden ReLU unit h = ReLU(x) and an **output** y = vh. The hidden **input** weight is fixed at 1, its bias is fixed at 0, and the **output** bias is fixed at 0. Only **output** weight v is trained.

This is a restricted two-layer network, not **training** every parameter of a general network.

Training pairs are (x, target) = (1, 2) and (2, 4). Both x values are positive, so hidden activations are 1 and 2. Define ordinary MSE over both rows.

Start at v = 0 and use learning rate 0.1 for exactly four full-data updates. The declared budget prevents stopping only when a preferred result appears.

All five recorded states for four updates

Update count	v	Training MSE
0	0	10
1	1	2.5

Update count	v	Training MSE
2	1.5	0.625
3	1.75	0.15625
4	1.875	0.0390625

Interpret the state and update counts

Each row evaluates the parameter value after that many updates. Row zero records the starting model. The loss decreases in this specific experiment; that pattern is not a promise for all networks or rates.

Name given and missing facts

> **List the given facts and one fact that is not supplied.**

MAIN LESSON · SECTION 3

Trace forward,loss,gradient and update through every state

> Five printed states do not mean five updates.

Derive the mean-loss gradient

Let errors be v - 2 and 2v - 4. MSE is their squared sum divided by 2, which simplifies to 2.5 times (v - 2) squared. Its gradient with respect to v is 5 times (v - 2).

At v = 0 the gradient is -10, so the first update gives v = 0 - 0.1 times -10 = 1.

Complete five-state training ledger

State after this many updates · v · Predictions on fixed h[1, 2] · Errors against[2, 4] · Training MSE · Gradient for next update · Next v

> **State after this many updates:** 0
>
> **v:** 0
>
> **Predictions on fixed h[1, 2]:** [0, 0]
>
> **Errors against[2, 4]:** [−2, −4]
>
> **Training MSE:** 10
>
> **Gradient for next update:** -10
>
> **Next v:** 1

> **State after this many updates:** 1
>
> **v:** 1
>
> **Predictions on fixed h[1, 2]:** [1, 2]
>
> **Errors against[2, 4]:** [−1, −2]
>
> **Training MSE:** 2.5
>
> **Gradient for next update:** -5
>
> **Next v:** 1.5

> **State after this many updates:** 2
>
> **v:** 1.5
>
> **Predictions on fixed h[1, 2]:** [1.5, 3]
>
> **Errors against[2, 4]:** [−.5, −1]
>
> **Training MSE:** 0.625
>
> **Gradient for next update:** -2.5
>
> **Next v:** 1.75

> **State after this many updates:** 3
>
> **v:** 1.75
>
> **Predictions on fixed h[1, 2]:** [1.75, 3.5]
>
> **Errors against[2, 4]:** [−.25, −.5]
>
> **Training MSE:** 0.15625
>
> **Gradient for next update:** -1.25
>
> **Next v:** 1.875

> **State after this many updates:** 4
>
> **v:** 1.875
>
> **Predictions on fixed h[1, 2]:** [1.875, 3.75]
>
> **Errors against[2, 4]:** [−.125, −.25]
>
> **Training MSE:** 0.0390625
>
> **Gradient for next update:** Stop:four-update budget
>
> **Next v:** No further update

Frozen-versus-trained value register

Value in this restricted network	Training status
Hidden input weight 1	Frozen
Hidden bias 0	Frozen
ReLU function	Fixed function, no fitted numerical parameter here
Output weight v	Only trainable value; start 0, end 1.875 after four updates
Output bias 0	Frozen
Data, rate.1, mean objective, four-update budget	Declared experiment configuration, not extra model parameters

Complete runnable four-update loop

```
train = [(1.0, 2.0), (2.0, 4.0)]
v = 0.0
rate = 0.1
for step in range(5):
    hs = [max(0.0, x) for x, target in train]
    errors = [v * h - target
              for h, (x, target) in zip(hs, train)]
    loss = sum(e * e for e in errors) / len(train)
    print(step, v, loss)
    if step == 4:
        break
    gradient = 2 * sum(e * h for e, h in zip(errors, hs)) / len(train)
    v = v - rate * gradient
```

The factor 2 comes from squared error; division by the number of rows makes this the gradient of the mean, not the sum. The code uses every **training** row for each update. It prints five states but performs only four updates.

All data are supplied fictional numbers.

Connect each visual step to the loop

Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

▼ **Earlier wording, alternatives and picture transcripts**

Connect the picture and the explanation

MAIN LESSON · SECTION 4

Change one stated condition and restart the experiment

> Do not mix a sum-loss gradient with the declared mean-loss objective.

Current mean-versus-sum task

- At v = 1, the sum of squared errors is 5. A learner uses gradient -10 with rate 0.1. Explain the mismatch with this lesson and give the next v under MSE.

> **Keep the procedure fixed. Explain what changed in the inputs and result.**

Predict the pictured target change

- Change: Change target from 3 to6 and restart v = 0 with the same rate.

▼ **Check after predicting**

Prediction: First gradient=-36; v becomes 2; **prediction** 6 and loss 0.

Reason: The hidden value remains 3, but the target residual and required **output** weight double.

▼ **Earlier wording, alternatives and picture transcripts**

Compare a familiar case with a changed case

Change one thing in the pictured case

MAIN LESSON · SECTION 6

See which value is fitted and which computation stays frozen

> The loss changes because v changes;the hidden representation does not learn here.

Complete pictured one-pair facts and scope

Train a single **output** weight while clearly marking the frozen hidden computation.

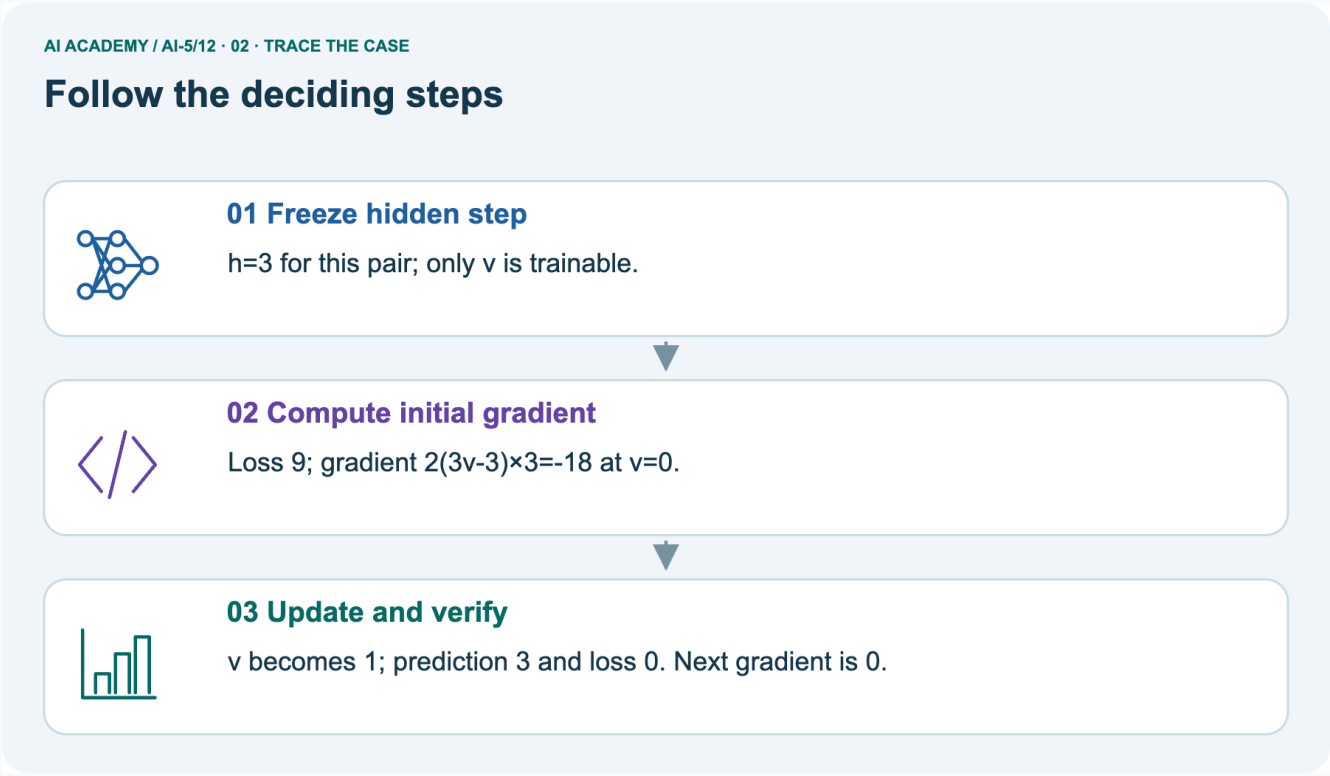
Toy h=ReLU(x), y=vh. One **training** pair x = 3, target 3; start v = 0; squared loss; rate 1/18.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

The picture is a separate one-pair experiment:x3, target 3, h3, start v0, squared error, rate 1/18. Initial gradient−18 gives v1, prediction 3, loss 0; the next gradient is0. Changing only its target to6 and restarting v0 gives initial gradient−36, v2, prediction 6, loss 0. This specially chosen rate and one pair differ from the main two-row MSE experiment.

Exact fit here does not establish memorization, learned hidden features or general image-classifier performance.

Trace frozen h,gradient and fitted v



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Freeze hidden step	h=3 for this pair; only v is trainable.
2. Compute initial gradient	Loss 9; gradient 2(3v-3)×3=-18 at v=0.
3. Update and verify	v becomes 1; prediction 3 and loss 0. Next gradient is 0.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

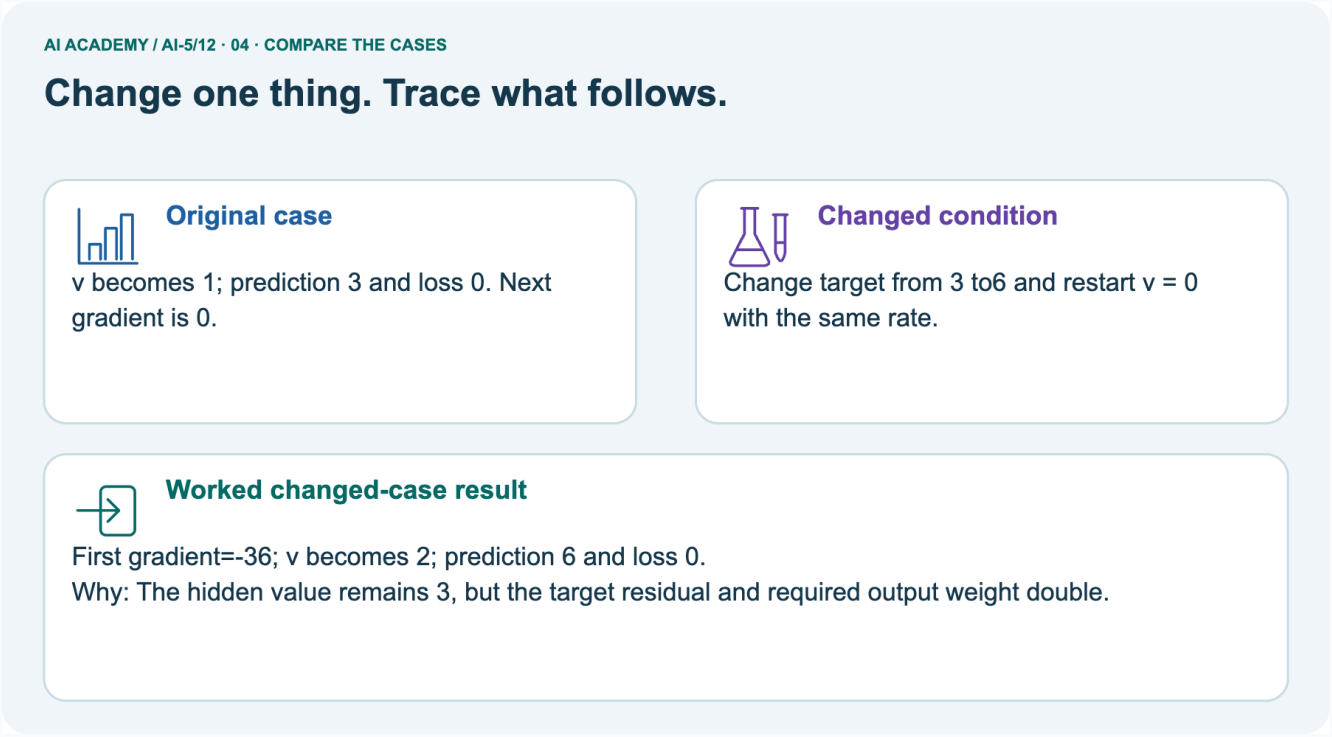
Only the **output** weight v is being fitted in this toy network; the hidden ReLU is fixed. The **training input** gives hidden value three.

Squared-error slope points v from zero towards one, and this specially chosen rate reaches one in a single update. Zero loss on one **training** pair is exact **fitting** of that pair, not **evidence** about other **input**s.

Predict the restarted doubled target

- Change one thing: Change target from 3 to6 and restart v = 0 with the same rate. Predict the result before checking the explanation. What remains the same?

▼ Check after predicting



Follow the labelled picture with the matching explanation. Open the vector diagram

Explain the trained value and evidence limits

- Zero loss on one example is an exact fit to that recorded example. It does not by itself establish memorisation or a generally useful image classifier.

- Read the labelled picture: Which parameter changed, and which part was not trained? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Try the main v1.5update

One full-data update from v is 1

STAGE	COMPUTE WITH CURRENT V	RESULT
Forward and loss	Predictions [1, 2] for targets [2, 4]	Errors [-1, -2], MSE 2.5
Gradient	2 times (-1 times 1 - 2 times 2) / 2	Gradient -5
Update with rate 0.1	1 - 0.1 times -5	Next v is 1.5

Each row depends on the preceding calculation. Recompute predictions and loss on the next iteration; do not keep using the first gradient after parameters change.

Starting from v = 1.5, calculate both training predictions, errors, MSE, gradient and the next v.

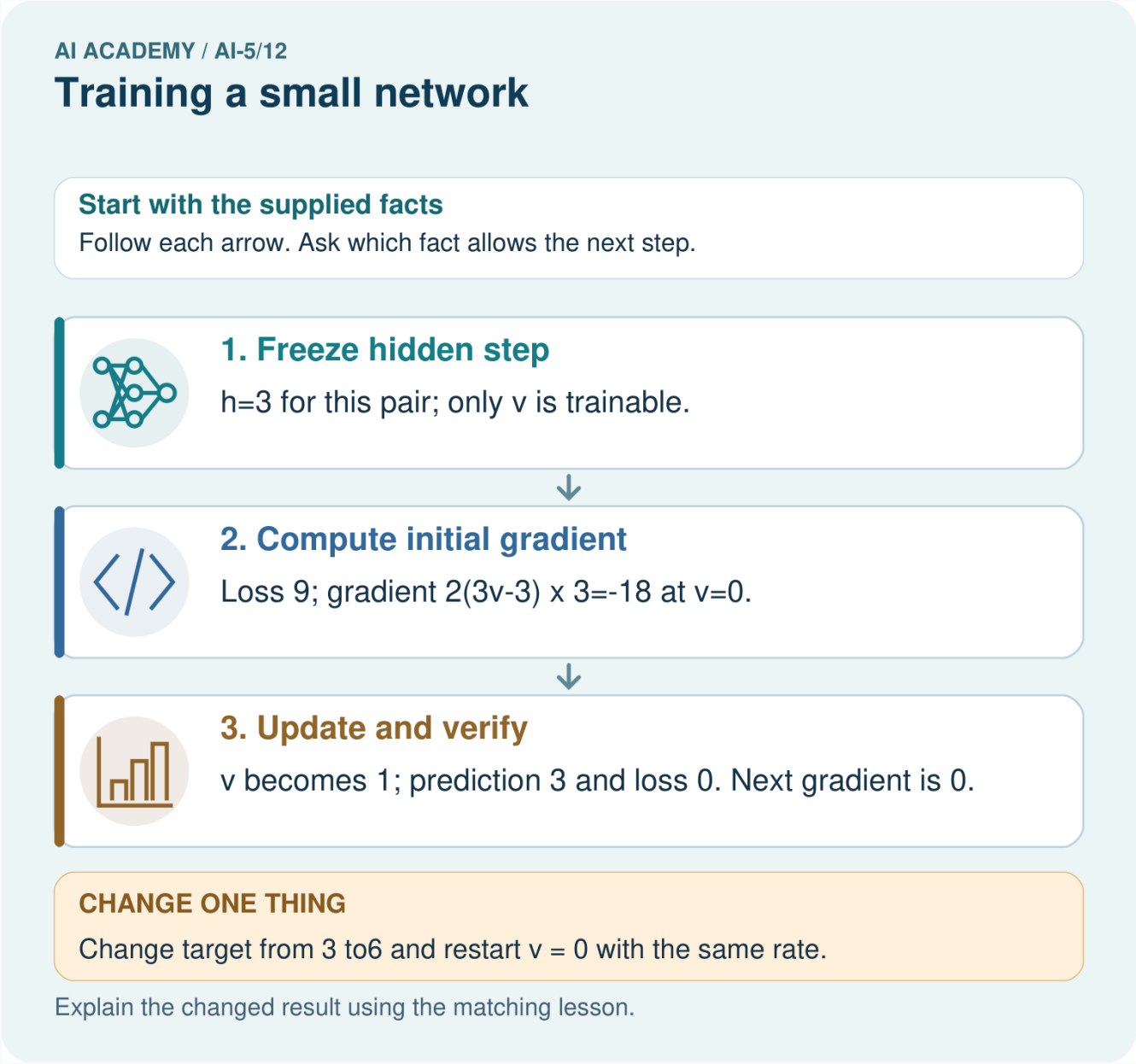
▼ **Earlier wording, alternatives and picture transcripts**

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Training a small network

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

MAIN LESSON · SECTION 7

Keep training evidence separate from the reserved demonstration

> A fixed hidden representation can make some distinctions impossible.

Complete reserved-row and negative-input explanations

After four updates, freeze v = 1.875. A reserved demonstration pair (3, 6), not used in the updates, receives **prediction** 5.625 and squared error 0.140625. Report it separately from **training** MSE 0.0390625.

One synthetic held-out example is a small demonstration, not **evidence** of broad real-world reliability.

The frozen representation has a limit: for any negative x, ReLU gives h = 0 and the **output** is always 0.

If the intended task were y = 2x for all real **input**s, x = -1 would need target -2, so this model would fail there regardless of v. **Training** cannot recover distinctions that the fixed hidden representation removes.

Complete post-freeze paired checks

After freezing main v1.875 · h · Prediction · Target · Squared error / status

> **After freezing main v1.875:** Training row x1
>
> **h:** 1
>
> **Prediction:** 1.875
>
> **Target:** 2
>
> **Squared error / status:** 0.015625

> **After freezing main v1.875:** Training row x2
>
> **h:** 2
>
> **Prediction:** 3.75

Target: 4

Squared error / status: 0.0625

After freezing main v1.875: Reserved demonstration x3

h: 3

Prediction: 5.625

Target: 6

Squared error / status: .140625; not used in the four updates

After freezing main v1.875: Representation probe x–1 under proposed y2 xrule

h: 0

Prediction: 0

Target: -2

Squared error / status: 4 for any v; structural counterexample, not another training row

Try the reserved-row arithmetic and split rule

- At final v = 1.875, calculate the **prediction** and squared error on reserved pair (3, 6). May its target be included in the four reported **training** updates?

State one repair and the checks you would rerun.

▼ Earlier wording, alternatives and picture transcripts

Locate the earliest unsupported step

MAIN LESSON · SECTION 8

Practise repeated-state reasoning with less help

> Name the objective,gradient and exact update count.

Current task and response

- At v = 0, calculate both **prediction**s, errors, MSE and gradient, then update at rate 0.1.

Starting from v = 1.5, calculate both training predictions, errors, MSE, gradient and the next v.

Explain your prediction to a partner before comparing results.

Supported W2record

Use workbook W2 for the connected example “**Training** a small network”. First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Individual misconception check

- Use the worked case for **Training** a small network. Proposed explanation: “Every parameter was trained”. Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

- Answer on your own before discussion. If you are unsure, say which step you need help with.

Pictured frozen-versus-trained reasoning

- Which parameter changed, and which part was not trained? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

> **My explanation using a pictured step and the given facts**

Reserve the independent new training pair

This is supported practice. Keep the separately named independent case for an attempt before feedback.

▼ Earlier wording, alternatives and picture transcripts

Practise with less help: Check the mean convention

Check your reasoning

Explain the picture

MAIN LESSON · SECTION 9

Run the complete restricted-network training experiment

> All data,initialization,loss,rate,gradients and stopping budget are supplied.

Current objective-scaling task and complete loop

At v = 1, the sum of squared errors is 5. A learner uses gradient -10 with rate 0.1. Explain the mismatch with this lesson and give the next v under MSE.

```
train = [(1.0, 2.0), (2.0, 4.0)]
v = 0.0
rate = 0.1
for step in range(5):
    hs = [max(0.0, x) for x, target in train]
    errors = [v * h - target
              for h, (x, target) in zip(hs, train)]
    loss = sum(e * e for e in errors) / len(train)
    print(step, v, loss)
    if step == 4:
        break
    gradient = 2 * sum(e * h for e, h in zip(errors, hs)) / len(train)
    v = v - rate * gradient
```

Complete normal,zero,missing,held-out and budget probes

Complete loop probe	Expected result / action
Normal declared two-row training	Four updates end v1.875, training MSE.0390625
At v1:sum versus mean	SSE 5, MSE 2.5, sum-gradient–10, mean-gradient–5; mean update gives v1.5
At v2on the training rows	Both predictions exact, gradient 0, no parameter change
Hidden zero boundary x0	h0, prediction 0 for any finite v; requires a target before computing loss
Missing target in a training row	Outside supplied complete-pair contract; request/review before fitting, not guess 0
Reserved pair(3, 6)	Score only after freeze; do not include in the four claimed updates
Negative x–1 under y2 xproposal	h0, prediction 0, target–2, error 4 regardless of v
Extra update after state 4	Changes the declared training budget; version and report it rather than call it the original run

Predict,run or trace,and retain all states

1. Predict all five states and four gradients before running or tracing the complete loop. Compare every v, prediction, error and loss against the supplied ledger; retain actual stdout or mark paper trace. Identify every frozen value. The code uses the two supplied complete numeric pairs and no missing-data validator. Keep reserved-row scoring outside fitting.

   A revised budget, objective, rate or hidden computation is a new configuration. Do not call one exact training fit either proved memorization or proved generalization.

2. On paper, trace the supplied procedure. If code is provided, predict before running it in a local practice file. Never write an expected answer as though it were an observed run.

Keep the evidence ledger

Record	Your evidence
Given input and fixed rule	
Prediction before checking	
Actual result, or “not run”	
First mismatch and one explanation	
Boundary or missing input and expected response	

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

▼ **Earlier wording, alternatives and picture transcripts**

Predict, investigate and preserve the evidence

Plan → try → check

MAIN LESSON · SECTION 14

Try alone: a different one-pair training experiment

> Use its target,rate and two-update budget rather than the worked experiment’s numbers.

Complete new independent training task

Keep h = ReLU(x), train only v, and use one pair (2, 2), v = 0 and rate 0.125. Use squared error and gradient 2 times (2v - 2) times 2. Perform two updates, reporting each new v and loss.

Explain why the second update changes nothing.

First attempt and support record

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Use feedback after trying

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

▼ Earlier wording, alternatives and picture transcripts

Independent case: Run a new training case independently

Attempt this without hints

Keep h = ReLU(x), train only v, and use one pair (2, 2), v = 0 and rate 0.125. Use squared error and gradient 2 times (2v - 2) times 2. Perform two updates, reporting each new v and loss.

Explain why the second update changes nothing.

My independent procedure, result and limitation

Record support used: none, hint or teacher explanation. After feedback, a different case is needed to demonstrate independence.

Use feedback to improve

MAIN LESSON · SECTION 11

Save the exact restricted experiment configuration

> A reader should know which parameter learned and which values stayed fixed.

Project artifact and evidence table

Add a labelled **training** a small network **evidence** sheet to Image-pattern Network Investigation. Show how you can trace repeated forward loss gradient and update steps.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

Project evidence	Record
Component and version	
Inputs permitted at prediction time	
Trace and comparison with earlier work	
Failure, limit and human review	

Actual run/trace and limits

For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

▼ Earlier wording, alternatives and picture transcripts

Evidence for your project

MAIN LESSON · SECTION 16

Finish with one complete cycle and one representation limit

> Distinguish a training state from an evaluation claim.

Retrieve and explain

- Close your notes. Explain this goal in your own words: Trace repeated forward loss gradient and update steps

My deciding step, evidence and one remaining question

Plan a changed-case return

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Use the mathematical words precisely**

REFERENCE · SECTION 5

Reference · Use the mathematical words precisely

> Connect each term to a value or step in the current example.

Vocabulary and your explanation

Word	Meaning
Training loop	A repeated sequence of prediction, loss calculation, gradient calculation and parameter update
Frozen parameter	A model value intentionally held fixed during fitting
Training budget	A declared limit on updates or other training resources
Learning curve	A record of a performance measure across training steps or time

Use two terms to explain a step in this week’s example.

▼ Extra challenge · Explain a stronger case

REFERENCE · SECTION 10

Extra challenge · Explain a stronger case

> Choose the next challenge after the main route.

Explain,repair and extend

- At v = 0, calculate both **prediction**s, errors, MSE and gradient, then update at rate 0.1.
- At final v = 1.875, calculate the **prediction** and squared error on reserved pair (3, 6). May its target be included in the four reported **training** updates?
- For the lesson architecture at x = -1, give h and **prediction** for any finite v. If target is -2, calculate squared error and explain why changing v cannot fix it.

Explain → apply → challenge

> **Explain which evidence supports your conclusion and which claim remains unproved.**

▼ Reference · Check the representation and its limits

REFERENCE · SECTION 12

Reference · Check the representation and its limits

> Use after practice;reserve the new independent one-pair two-update training task for the first individual attempt.

Current explanation and response

- Save the model formulas, frozen values, trained parameter, initial value, data pairs, loss definition, learning rate, update budget and the printed trace. State that these numbers are a classroom experiment.

  Changing a setting after seeing reserved outcomes makes those outcomes development **evidence** for that revision.

- Next week separates an update, a batch and an epoch when **training** uses more than one group of examples.

- For the lesson architecture at x = -1, give h and **prediction** for any finite v. If target is -2, calculate squared error and explain why changing v cannot fix it.

A team reports that its training MSE fell each step and concludes the network learned y = 2x for every real x. Give one numerical counterexample and a more accurate conclusion.

My limited claim and a useful next check

▼ Extra practice · Revisit the complete trace

REFERENCE · SECTION 13

Extra practice · Revisit the complete trace

> Predict before reading the supplied steps.

Complete trace and evidence

Complete evidence record

One full-data update from v is 1

STAGE	COMPUTE WITH CURRENT V	RESULT
Forward and loss	Predictions [1, 2] for targets [2, 4]	Errors [-1, -2], MSE 2.5
Gradient	2 times (-1 times 1 - 2 times 2) / 2	Gradient -5
Update with rate 0.1	1 - 0.1 times -5	Next v is 1.5

Each row depends on the preceding calculation. Recompute predictions and loss on the next iteration; do not keep using the first gradient after parameters change.

Follow the **input** through the stated procedure to its result. Cover the result first, predict it, then explain every intermediate step. The text/table accompanies the figure so the example can be read without colour.

My complete trace of the worked case

▼ Reference · The earlier route checklist

REFERENCE · SECTION 15

Reference · The earlier route checklist

> Use one main route rather than repeating the lesson.

Targets,route and first prediction

Start with this question: Train a single **output** weight while clearly marking the frozen hidden computation.

By the end, I can	How I will show it
Trace repeated forward loss gradient and update steps	Explain the deciding step, show a trace or test, and state one limit.
Keep frozen parameters distinct from the trained parameter	Explain the deciding step, show a trace or test, and state one limit.
Report training results with their evaluation limits	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

▼ **Extra practice · Predict,check and repair**

REFERENCE · SECTION 17

Extra practice · Predict,check and repair

> Keep supported feedback separate from the independent case.

First supported question and optional hint

- Train a single **output** weight while clearly marking the frozen hidden computation.
- Which labelled step matches this detail? v becomes 1; **prediction** 3 and loss 0. Next gradient is 0.
- 1. Freeze hidden step
- 2. Update and verify
- 3. Compute initial gradient
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check after the first attempt**

After your attempt, compare with the supported explanation: Update and verify v becomes 1; **prediction** 3 and loss 0. Next gradient is 0. Only the **output** weight v is being fitted in this toy network; the hidden ReLU is fixed.

The **training input** gives hidden value three. Squared-error slope points v from zero towards one, and this specially chosen rate reaches one in a single update.

Zero loss on one **training** pair is exact **fitting** of that pair, not **evidence** about other **input**s.

Changed-case question and optional hint

- Changed-case practice: Change target from 3 to6 and restart v = 0 with the same rate. Which conclusion is supported by the supplied facts?
- 1. First gradient=-36; v becomes 2; **prediction** 6 and loss 0.
- 2. We can decide the result without examining the changed **input** or the procedure.
- 3. This one example guarantees correct results for every future **input**.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check after the changed attempt**

After your attempt, compare with the supported explanation: First gradient=-36; v becomes 2; **prediction** 6 and loss 0. The hidden value remains 3, but the target residual and required **output** weight double.

Repair and retain the record

Review Updating parameters (ai-5/11). Goal: Apply a learning rate to an ordered gradient. Short practice: Why must new loss be recomputed rather than assumed smaller?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Slopes and local change (ai-5/9). Goal: Calculate an average rate of change over a stated interval. Short practice: Which calculation disproves “any increase helps”?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Forward passes by hand (ai-5/6). Goal: Record scores and activations in dependency order. Short practice: Why should feedback identify h before correcting only y?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

▼ Earlier wording, alternatives and picture transcripts

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Predict, run and debug

Keep your completed work

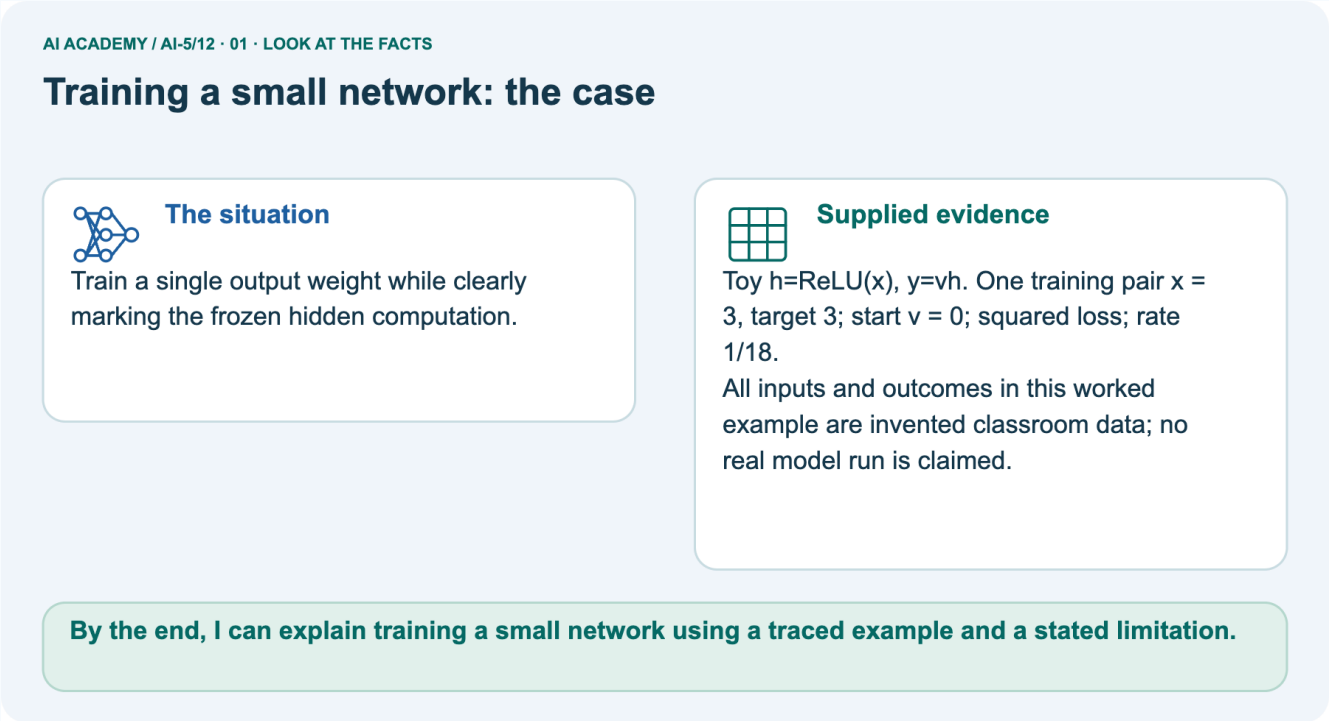
▼ Reference · Six-picture concept journey

REFERENCE · SECTION 18

Reference · Six-picture concept journey

> Choose the picture that supports the current explanation.

01. Training a small network: the case



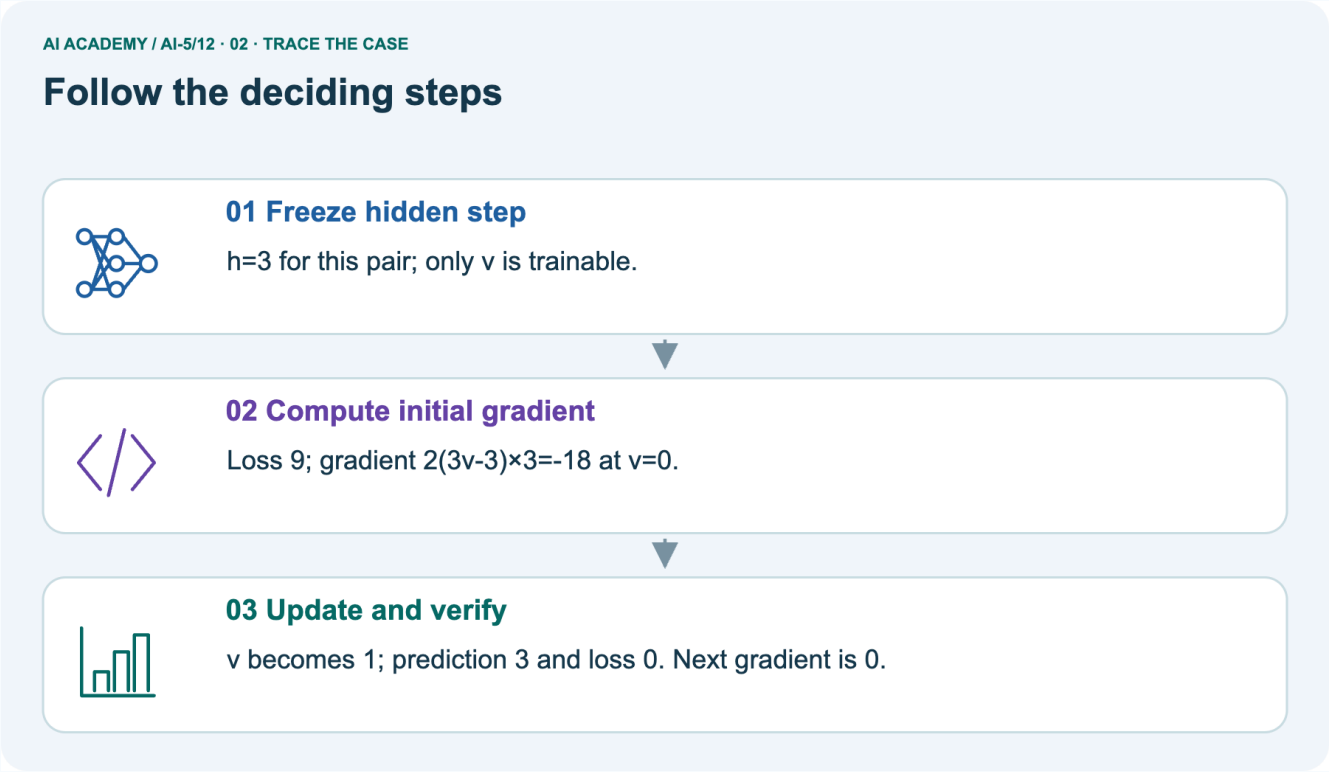
Follow the labelled picture with the matching explanation. Open the vector diagram

Train a single **output** weight while clearly marking the frozen hidden computation. Supplied facts: Toy h=ReLU(x), y=vh. One **training** pair x = 3, target 3; start v = 0; squared loss; rate 1/18.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps

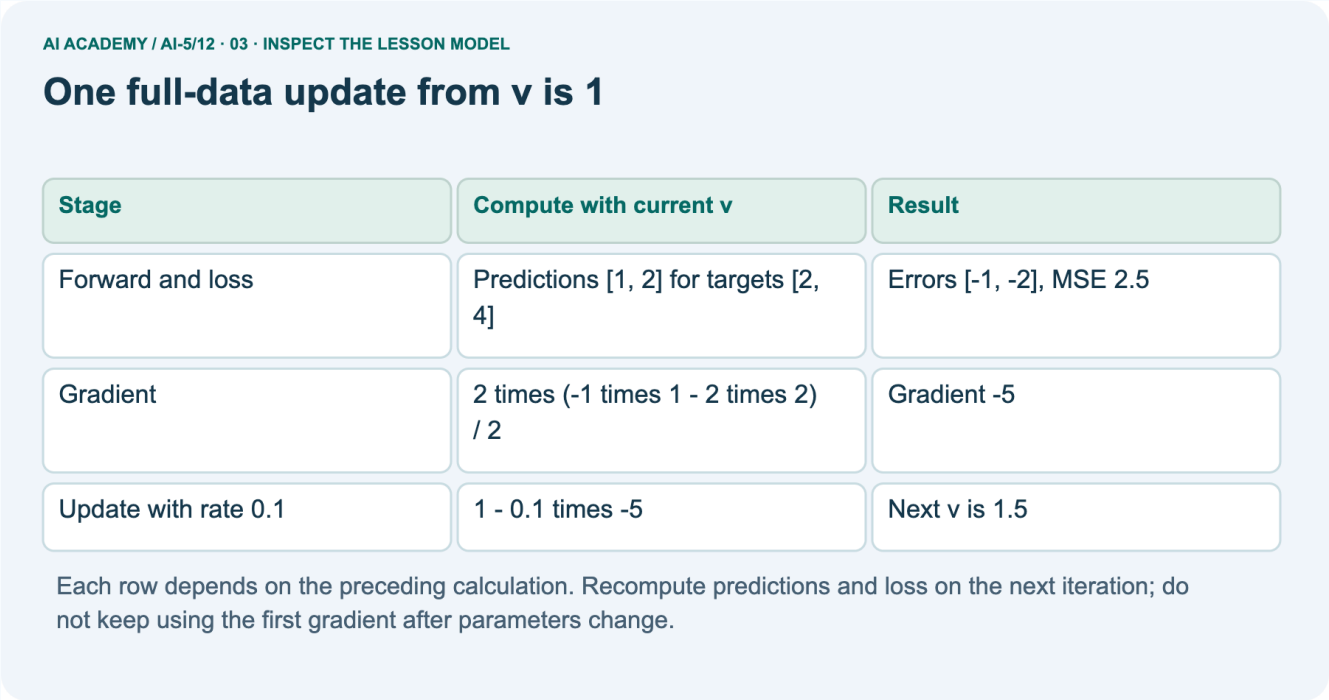


Follow the labelled picture with the matching explanation. Open the vector diagram

Freeze hidden step: h=3 for this pair; only v is trainable. → Compute initial gradient: Loss 9; gradient 2(3v-3)×3=-18 at v=0. → Update and verify: v becomes 1; **prediction** 3 and loss 0. Next gradient is 0.

Point to each numbered stage and explain which recorded **input** it uses.

03. One full-data update from v is 1



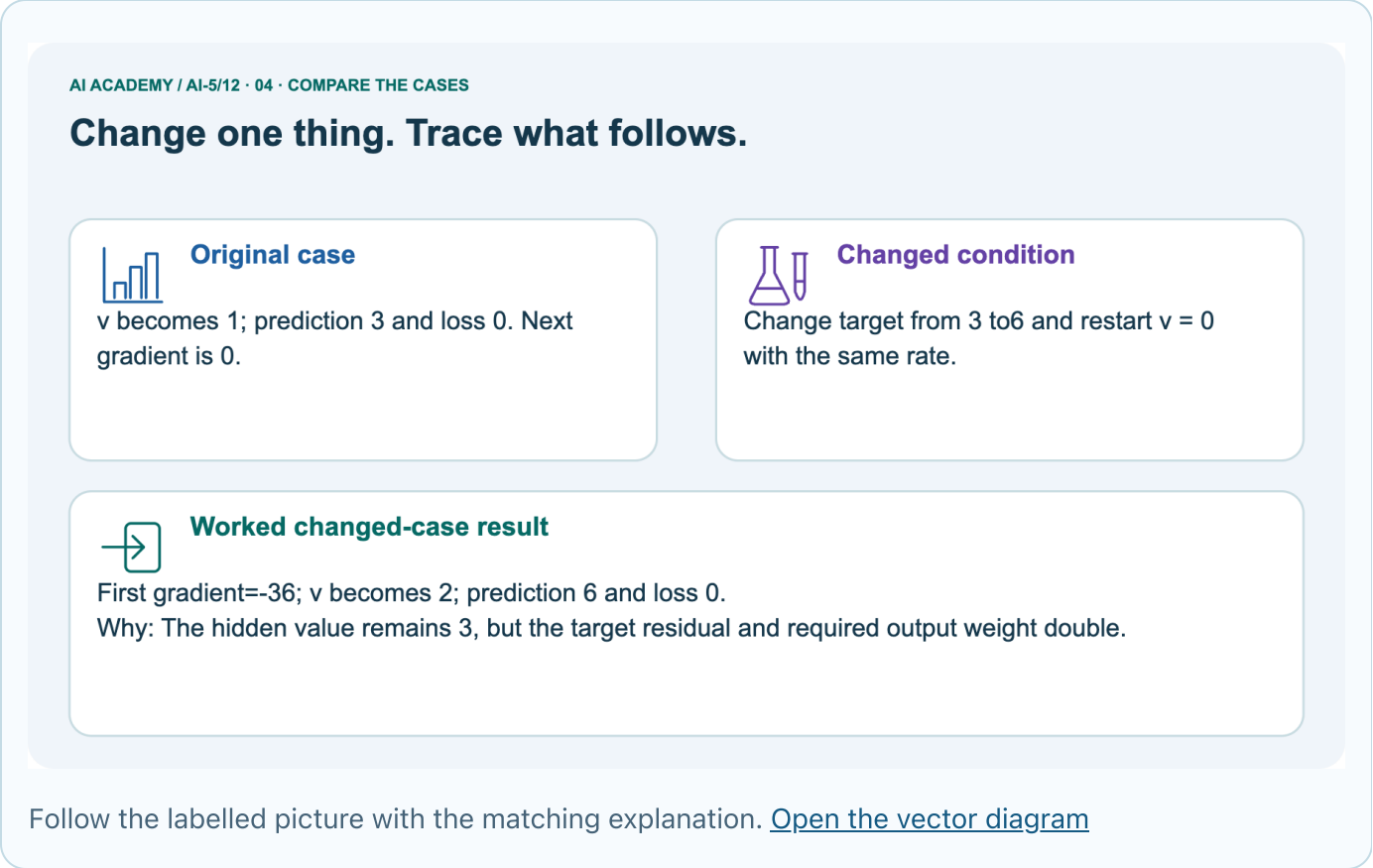
Follow the labelled picture with the matching explanation. Open the vector diagram

One full-data update from v is 1. Forward and loss | **Prediction**s [1, 2] for targets [2, 4] | Errors [-1, -2], MSE 2.5. Gradient | 2 times (-1 times 1 - 2 times 2) / 2 | Gradient -5.

Update with rate 0.1 | 1 - 0.1 times -5 | Next v is 1.5

Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



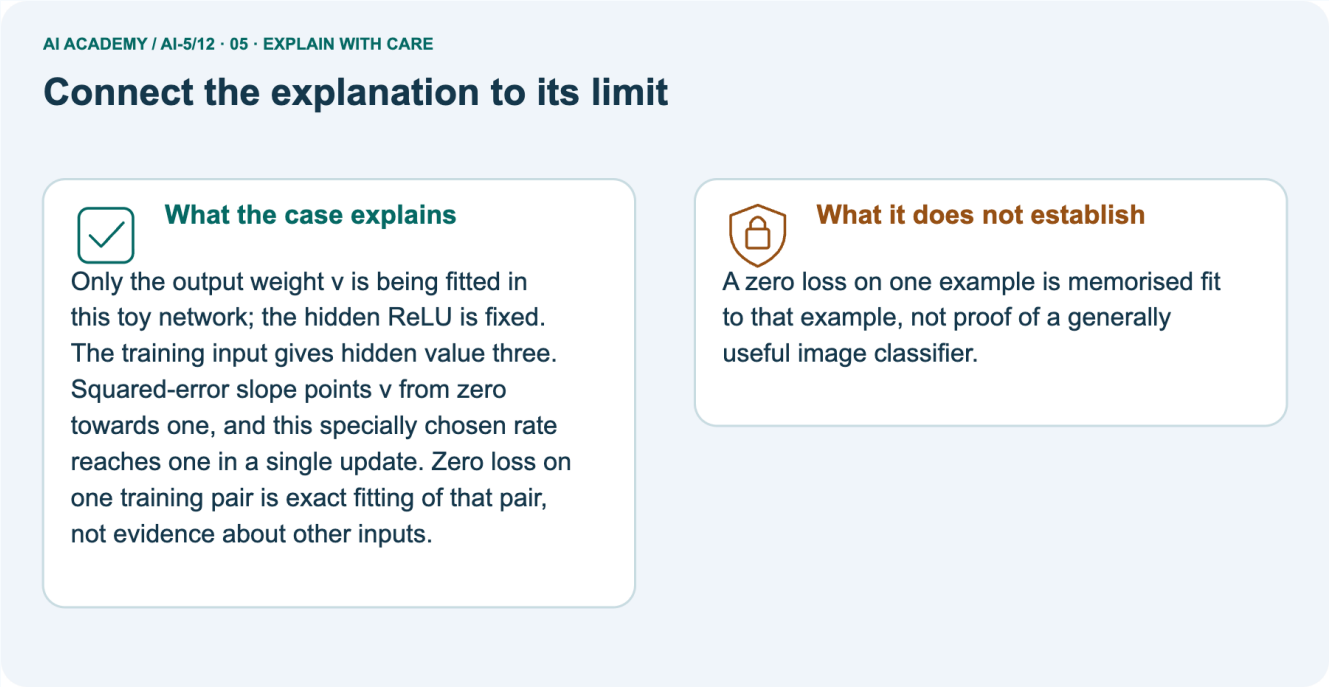
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: v becomes 1; **prediction** 3 and loss 0. Next gradient is 0. Changed condition: Change target from 3 to6 and restart v = 0 with the same rate.

Expected result: First gradient=-36; v becomes 2; **prediction** 6 and loss 0. Reason: The hidden value remains 3, but the target residual and required **output** weight double.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit

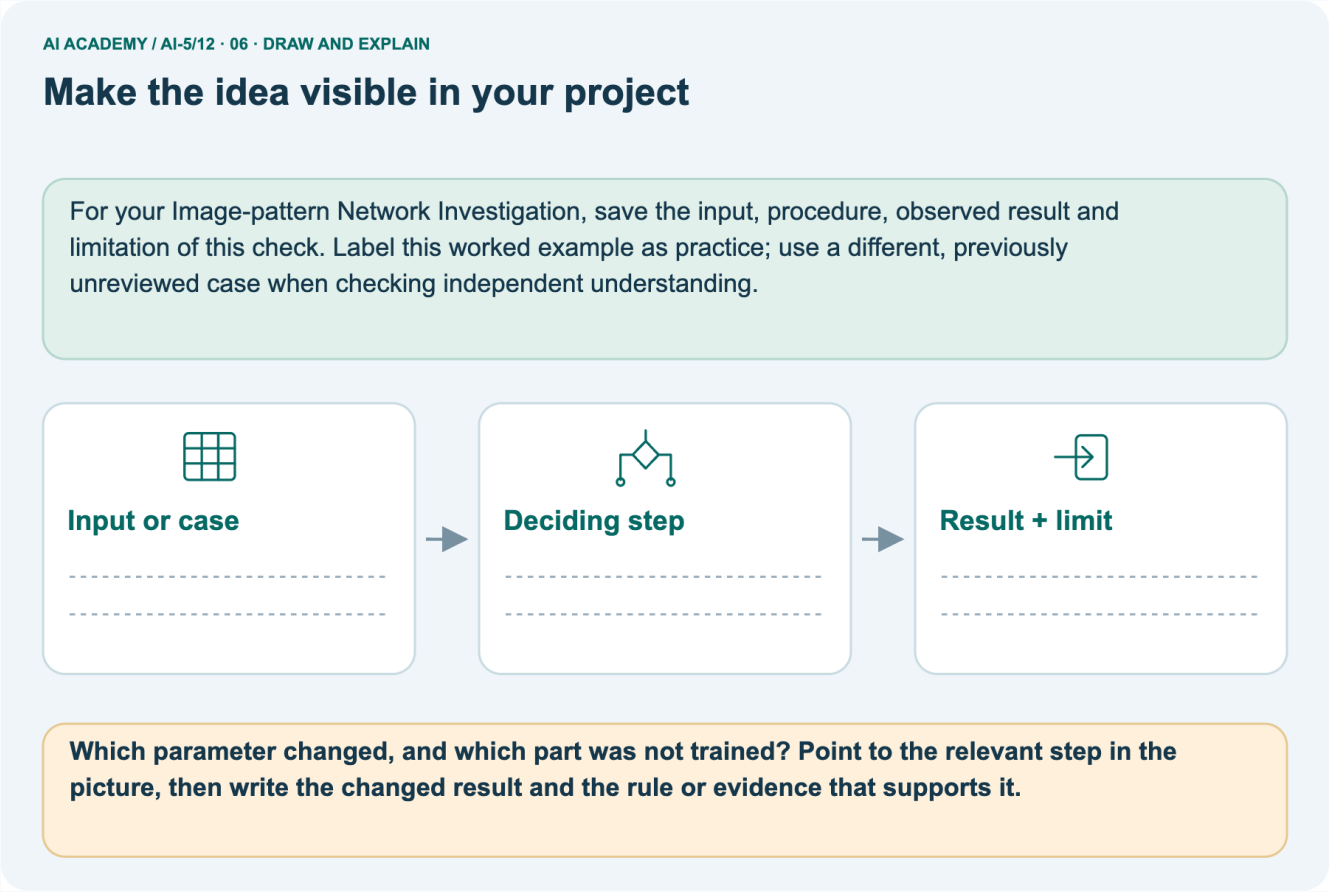


Follow the labelled picture with the matching explanation. Open the vector diagram

Only the **output** weight v is being fitted in this toy network; the hidden ReLU is fixed. The **training input** gives hidden value three. Squared-error slope points v from zero towards one, and this specially chosen rate reaches one in a single update. Zero loss on one **training** pair is exact **fitting** of that pair, not **evidence** about other **input**s. Limit: Zero loss on one example is an exact fit to that recorded example. It does not by itself establish memorisation or a generally useful image classifier.

Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Which parameter changed, and which part was not trained? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Draw your project trace and justify the deciding step using **evidence**.

▼ Earlier wording, alternatives and picture transcripts

01. Training a small network: the case

02. Follow the deciding steps

03. One full-data update from v is 1

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.